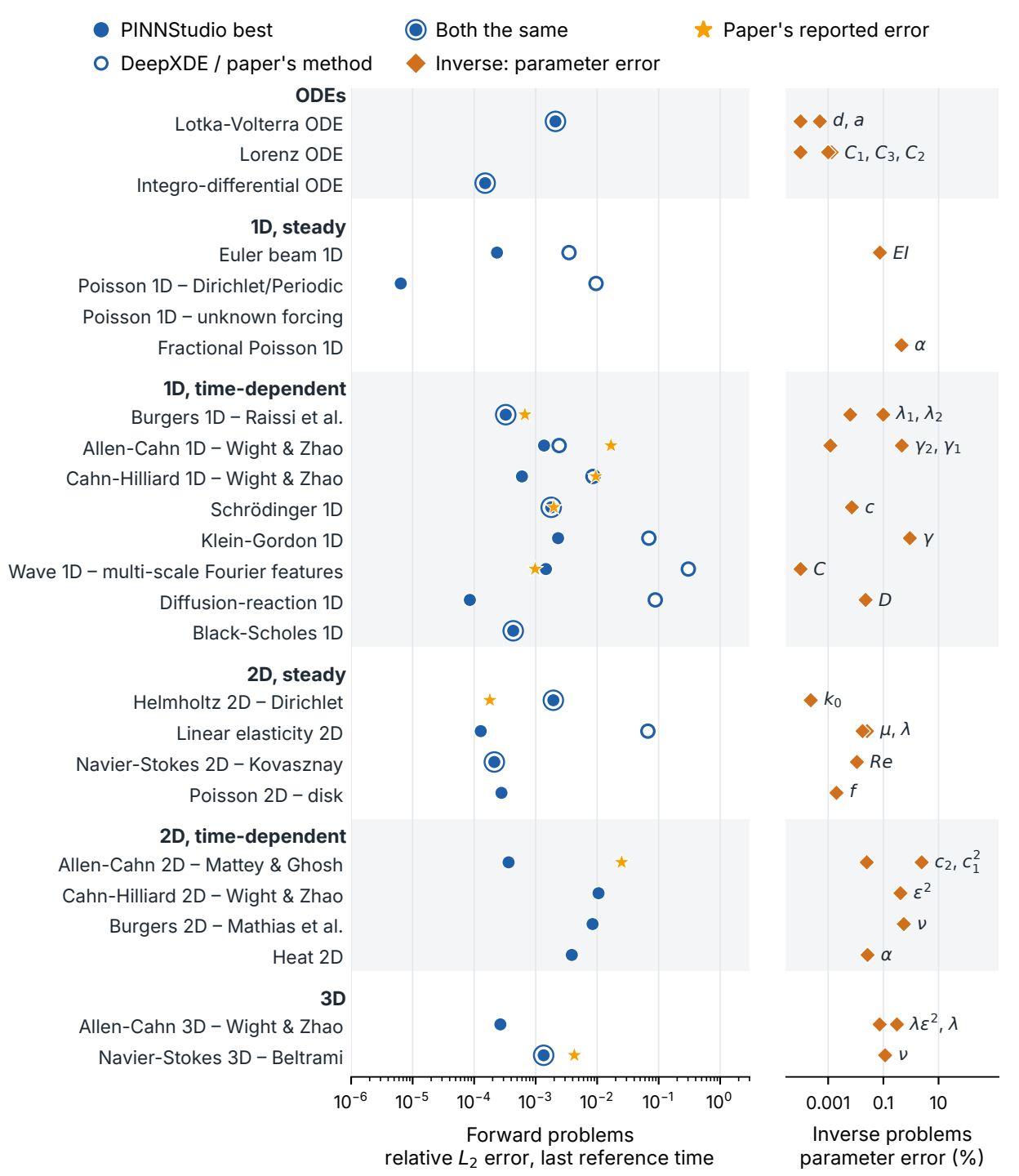
PINNStudio best
Both the same
Paper's reported error
DeepXDE / paper's method
Inverse: parameter error
ODEs
Lotka-Volterra ODE
Lorenz ODE
Integro-differential ODE
d, a
C1, C3, C2
1D, steady
Euler beam 1D
Poisson 1D – Dirichlet/Periodic
Poisson 1D – unknown forcing
Fractional Poisson 1D
EI
α
1D, time-dependent
Burgers 1D – Raissi et al.
Allen-Cahn 1D – Wight & Zhao
Cahn-Hilliard 1D – Wight & Zhao
Schrödinger 1D
Klein-Gordon 1D
Wave 1D – multi-scale Fourier features
Diffusion-reaction 1D
Black-Scholes 1D
λ1, λ2
γ2, γ1
c
γ
C
D
2D, steady
Helmholtz 2D – Dirichlet
Linear elasticity 2D
Navier-Stokes 2D – Kovasznay
Poisson 2D – disk
k0
μ, λ
Re
f
2D, time-dependent
Allen-Cahn 2D – Mattey & Ghosh
Cahn-Hilliard 2D – Wight & Zhao
Burgers 2D – Mathias et al.
Heat 2D
c2, c1^2
ε^2
ν
α
3D
Allen-Cahn 3D – Wight & Zhao
Navier-Stokes 3D – Beltrami
λε^2, λ
ν
10^-6 10^-5 10^-4 10^-3 10^-2 10^-1 10^0
Forward problems
relative L2 error, last reference time
0.001 0.1 10
Inverse problems
parameter error (%)